

## Project Executable Files

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 24 June 2026                                                |
| <b>Team ID</b>       | SWTID-2026-3249                                             |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 3 Marks                                                     |

### Project Executable Files

This section requires submission of all executable and deployable files for the OptiCrop project. All files are organized, named, and uploaded to the specified submission platform so the evaluator can run the solution using the provided files and instructions.

#### Step 1: Submission Checklist

| S.No | Item to Submit                                           | Submitted (Yes/No/NA)                                     |
|------|----------------------------------------------------------|-----------------------------------------------------------|
| 1    | Complete source code (all files and folders)             | Yes                                                       |
| 2    | README / Setup Guide (instructions to run the project)   | Yes                                                       |
| 3    | requirements.txt / dependency file                       | Yes                                                       |
| 4    | Database schema / seed files (if applicable)             | NA - uses Crop_recommendation.csv as the training dataset |
| 5    | Environment configuration file (.env.example or similar) | NA - no environment secrets required                      |
| 6    | Deployed application URL (if hosted)                     | No - runs locally via Flask development server            |
| 7    | APK / Executable binary (if applicable)                  | NA                                                        |
| 8    | Dockerfile / Containerization config (if applicable)     | No                                                        |
| 9    | Test files and test results                              | Yes - manual and performance test results included        |
| 10   | Demo video or walkthrough (if required)                  | Yes                                                       |

#### Step 2: File / Folder Structure

Overview of the OptiCrop project's file and folder structure:

```

opticrop/
  app.py - Flask application and prediction routes
  train.py - Model training and preprocessing pipeline
  model.pkl - Serialized trained classification model
  Crop_recommendation.csv - Training dataset
  requirements.txt - Python dependencies
  README.md - Setup and run instructions
  templates/ - Jinja2 HTML templates (layout, index, predict, about)
  static/css/ - Stylesheet (style.css)
  
```

#### Step 3: Deployment / Access Details

| Field                 | Details                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL | Not yet hosted - currently runs on the local Flask development server ( <a href="http://127.0.0.1:5000">http://127.0.0.1:5000</a> ) |

| Field                       | Details                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------|
| Login Credentials (Demo)    | Not applicable - the application has no authentication layer                                            |
| Platform / Hosting Provider | Planned: Heroku / AWS (see Future Scope in project documentation)                                       |
| Repository Link             | <a href="https://github.com/GeethikaBeduduri/OptiCrop">https://github.com/GeethikaBeduduri/OptiCrop</a> |
| Demo Video Link             | To be added upon recording                                                                              |

#### Step 4: Run Instructions

**Pre-requisites:** Python 3.9+, pip, and the packages listed in requirements.txt (Flask, NumPy, Pandas, Scikit-learn)

Step 1: Clone the repository: `git clone https://github.com/GeethikaBeduduri/OptiCrop`

Step 2: Install dependencies: `pip install -r requirements.txt`

Step 3: (Optional) Retrain the model: `python train.py`

Step 4: Start the application: `python app.py`

Step 5: Open a browser at `http://127.0.0.1:5000`

#### Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                             | Workaround / Status                                    |
|------|----------------------------------------------------------------------|--------------------------------------------------------|
| 1    | Flask development server is single-threaded and not production-grade | Planned migration to Gunicorn/WSGI for deployment      |
| 2    | No real-time weather or IoT sensor integration yet                   | Listed as a future enhancement in the project roadmap  |
| 3    | Model limited to the 22 crop classes present in the training dataset | Can be retrained with an expanded dataset via train.py |